EXPERIMENT 4

AIM : **Serial communication (between controller boards or any other device)**.

THEORY :

Serial communication is a way of sending data between two devices, one bit at a time, through a communication line.It is often used to connect controller boards, sensors, computers, and other electronic devices.Unlike parallel communication, serial communication uses fewer wires, which makes the setup simpler, cheaper, and easier to build.The data is sent in a special order so that the receiving device can understand it correctly.

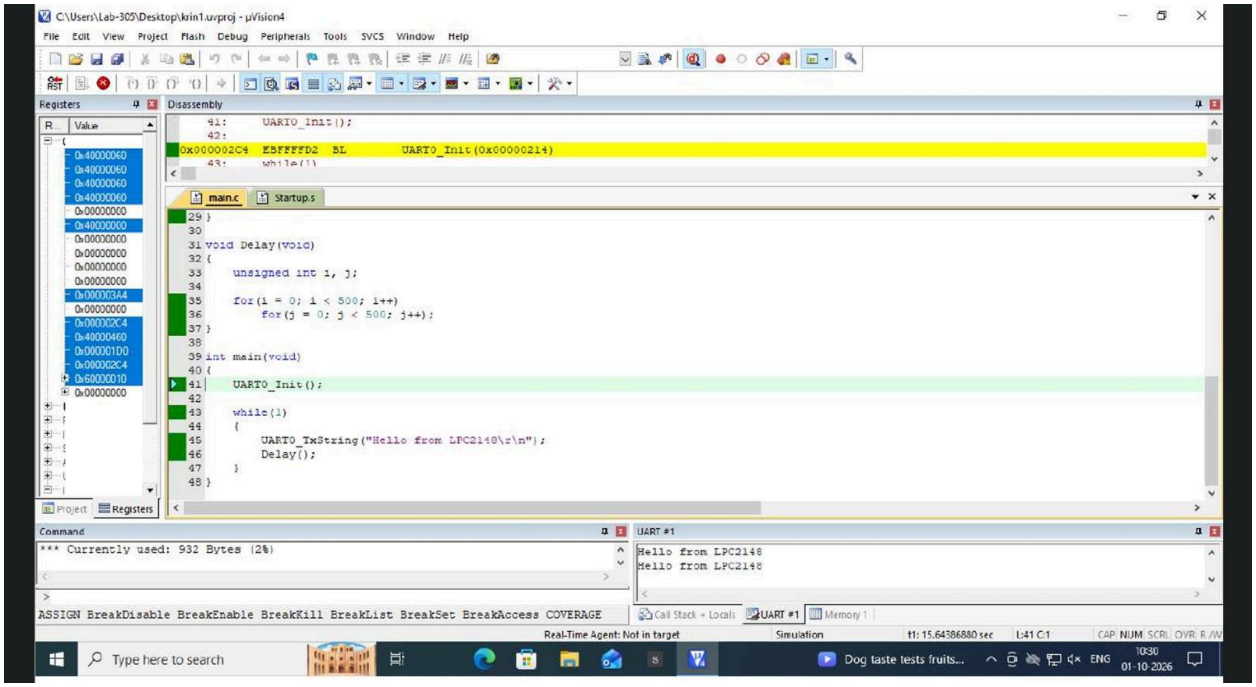
In a serial communication setup, one device sends the data and the other receives it.
The sending device turns the data into a series of bits and sends them through the wire.The receiving device collects these bits and puts them back together to get the original message.Some communication protocols use extra bits like start bits, stop bits, parity bits, or clock signals to keep both devices in sync and prevent errors.Common types of serial communication include UART, SPI, and IÂ²C, each with its own way of sending and matching data.

Serial communication is popular in embedded systems because it allows controllers to talk to other devices with very few connections.
For example, a microcontroller can send or receive data from another controller, a computer, a screen, a sensor, or a communication module using a serial interface.The speed of the communication, called the baud rate in some systems, needs to be set the same on both devices.This makes serial communication a useful and effective way for electronic devices to share information, and it's a key idea in working with microcontrollers and embedded systems.

ALGORITHM :

1. Begin the program.
2. Set up UART0 on the LPC2148 microcontroller by using the `UART0_Init()` function.
3. Create a Delay function with two nested `for` loops to make a time delay.
4. Enter a loop that runs forever using `while(1)`.
5. Send the message "Hello from LPC2148\r\n" through UART0 using `UART0_TxString()`.
6. Use the Delay function to pause between sending the message each time.
7. Keep repeating steps 5 and 6 over and over again.
8. The message "Hello from LPC2148" keeps showing up on the UART terminal.

OUTPUT :



CONCLUSION :

The experiment successfully demonstrated serial communication using the LPC2148 microcontroller and UART0 . The UART was initialized and the message "Hello from LPC2148" was transmitted repeatedly with a suitable time delay. The message was successfully received and displayed on the UART terminal. Thus, the experiment verified that UART-based serial communication can be effectively used to transmit data between a microcontroller and another device .